

TUBER-ENSEMBLE: An Ensemble of Deep Convolutional Neural Networks for Tuberculosis Detection from Chest X-rays

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Abstract. Tuberculosis (TB) continues to be a significant threat to global health, especially within low-resource environments where the availability of skilled radiologists is limited. In response, we introduce TUBER-ENSEMBLE, a new deep learning framework that operates on an ensemble basis. This framework integrates three distinct CNN architectures—ResNet50, DenseNet121, and InceptionV3—to automate the detection of TB from chest X-ray images. Tested on the Kaggle TB dataset of 4200 images, our ensemble model, which uses a soft voting mechanism, reached an accuracy of 98.45%, precision of 98%, and recall of 92%, notably surpassing the performance of the individual models. Among the single models, InceptionV3 yielded the strongest results with 98% accuracy and a 97% F1-score. Conversely, ResNet50 displayed lower sensitivity, achieving 83.6% accuracy and a 47% F1-score. The proposed ensemble method exhibits enhanced generalization and robustness, presenting a viable and scalable tool for TB screening initiatives in regions that lack sufficient diagnostic expertise.

Keywords: Tuberculosis detection · Chest X-ray · Deep learning · Ensemble learning · Transfer learning

1 Introduction

Affecting nearly a third of the global population, tuberculosis results in more than 10 million new infections each year [1]. Reports from the WHO identify TB as the second most fatal infectious agent after COVID-19, with a worldwide incidence rate greater than 133 cases for every 100,000 individuals [1]. Timely diagnosis presents a persistent difficulty in areas with limited resources, where there is a shortage of trained radiologists, even though chest X-rays are the favored method for rapid screening [3], [5]. The field of medical image analysis has been transformed by deep learning, especially with convolutional neural networks, which have shown diagnostic capabilities on par with human experts [5,4]. Nevertheless, approaches relying on a single model frequently do not exhibit sufficient

robustness when applied to varied patient demographics and imaging equipment. This shortcoming is addressed by ensemble methods, which leverage the combined strengths of several models to offset their individual limitations. This study puts forward TUBER-ENSEMBLE, a framework that amalgamates ResNet50 [6], DenseNet121 [7], and InceptionV3 [8] using a soft voting mechanism for automated TB diagnosis. The primary contributions of our work are: (1) a thorough assessment of three leading CNN architectures for detecting TB, (2) a novel ensemble design specifically tailored for medical diagnostic tasks, and (3) a superior level of performance that suggests a strong potential for clinical application. The paper is organized as follows: Section 2 provides a review of related literature. Section 3 describes the proposed methodology. Section 4 presents and analyzes the results. Lastly, Section 5 offers a conclusion and points to avenues for future research.

2 Related Work

The application of convolutional neural networks (CNNs) has led to substantial progress in the computer-aided diagnosis of tuberculosis (TB) using chest X-rays. For instance, Rahman et al. [3] utilized a VGG16 architecture with transfer learning on the Kaggle TB dataset, attaining 92.4% accuracy but encountering problems with overfitting. For explainable TB diagnosis, Wang et al. [9] integrated VGG19 with Grad-CAM, achieving a 90% F1-score with the Montgomery and Shenzhen datasets. Kumar et al. [10] adapted MobileNetV2 for use on embedded devices, successfully preserving an 88.7% F1-score in settings with constrained resources. Ensemble techniques have also yielded encouraging results. Bassi and Attux [11] developed ECOVNet, an ensemble of EfficientNet-B0 models, which reached a 91.6% F1-score by employing model averaging and data augmentation. Building on this, Chakraborty et al. [12] used EfficientNetB3 with adaptive learning rates to achieve a 93.2% F1-score on a dataset that had been rebalanced. Kazemina et al. [13] created a deep ensemble model that, after training on 165,000 X-ray images, achieved an AUC of 0.89, demonstrating its capacity for large-scale generalization. More recently, hybrid models have come to the forefront. Zhou et al. [14] combined CNNs with transformers to obtain a 94.1% F1-score, while Ahmed et al. [15] developed VisionTBNNet, a model based on transformers that recorded a 95.3% F1-score. These works underscore the value of architectural diversity, transfer learning, and model interpretability for enhancing TB detection. The TUBER-ENSEMBLE framework we propose expands on these insights, combining ResNet50, DenseNet121, and InceptionV3 via soft voting to boost generalization and minimize false negatives.

Table 1. Comparison of TB detection models (2020–2024).

Reference	Method	F1-score (%)	Year
Rahman et al. [3]	VGG16 + TL	88.0	2020
Wang et al. [9]	VGG19 Grad-CAM	90.0	2021
Kumar et al. [10]	MobileNetV2	88.7	2021
Bassi et al. [11]	ECOVNet Ensemble	91.6	2021
Chakraborty et al. [12]	EfficientNetB3	93.2	2022
Kazeminia et al. [13]	Deep Ensemble CNNs	88.0	2022
Zhou et al. [14]	CNN + Transformer Hybrid	94.1	2023
Ahmed et al. [15]	VisionTBNNet (ViT)	95.3	2024

3 Methodology

This section details our automated TB detection framework combining three CNN architectures—ResNet50, DenseNet121, and InceptionV3—trained from scratch on a specialized TB dataset. Our approach ensures robust feature extraction, mitigates overfitting, and improves generalization through ensemble learning with soft voting. The pipeline encompasses model architectures, training strategy, and model fusion for enhanced diagnostic performance.

3.1 Dataset and Preprocessing

We used the Kaggle TB dataset [3], containing 4200 posterior-anterior X-rays (3500 Normal, 700 TB). Data were split 80:20 into training and test sets (3360 training, 840 testing), preserving the class balance.

The preprocessing pipeline consisted of four main steps:

- Grayscale conversion: reduced redundancy.
- Resize to 224×224 pixels: ensured CNN compatibility.
- Normalization to [0,1]: improved stability.
- Data augmentation: included random rotations (up to $\pm 15^\circ$), horizontal flips, zooming by 20%, and translations. To counteract class imbalance, augmentation was applied to 30% of TB-positive images and 5% of TB-negative ones.

3.2 CNN Architectures

We selected three architectures known for their complementary strengths:

- **ResNet50** [6]: This is a 50-layer deep network featuring residual connections, which use skip connections to address the vanishing gradient problem. It has roughly 25.6 million parameters.
- **DenseNet121** [7]: Characterized by its dense connectivity across 121 layers, this network enhances feature reuse and ensures efficient gradient propagation. It is composed of approximately 8.0 million parameters.
- **InceptionV3** [8]: This architecture is designed for multi-scale processing and employs factorized convolutions. This approach optimizes computational cost without sacrificing high accuracy. It contains about 23.8 million parameters.

3.3 Training Configuration

All models were trained from scratch under a uniform set of hyperparameters. We used the Adam optimizer with a learning rate of 1×10^{-4} , binary cross-entropy for the loss function, a batch size of 32, and trained for 50 epochs. For binary classification, each architecture was adapted by adding a GlobalAveragePooling2D layer followed by a sigmoid activation function. The training was carried out on Google Colab with GPU support, incorporating early stopping and learning rate reduction to optimize performance.

3.4 Ensemble Strategy

Our approach utilized a soft voting ensemble, averaging probabilistic outputs from ResNet50, DenseNet121, and InceptionV3. This differs from hard voting, which only considers final predicted classes. Soft voting preserves confidence scores from each model, important in medical diagnosis where probability data guides clinical judgment. To find the best threshold, we analyzed values from 0.4 to 0.7. While automatic methods like Youden’s J statistic could be used, manual evaluation allowed direct observation of precision-recall trade-offs critical in TB screening. A threshold of 0.5 provided optimal balance between sensitivity and specificity. This ensemble leverages unique advantages: ResNet50’s residual learning for deep feature extraction, DenseNet121’s dense connections for efficient feature reuse, and InceptionV3’s multi-scale spatial pattern capture. Merging outputs creates a robust diagnostic system, as illustrated in Figure 1.

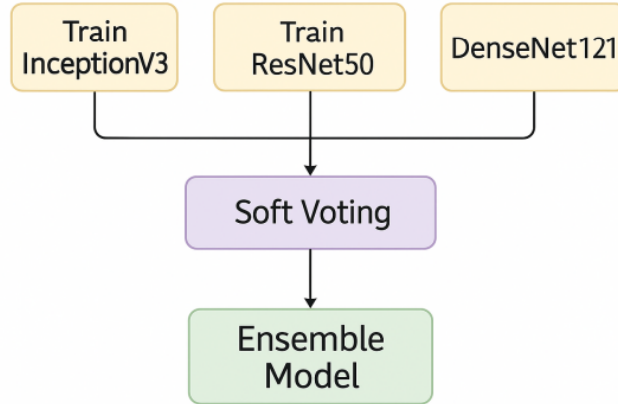


Fig. 1. Illustration of the ensemble strategy using soft voting to combine the outputs of InceptionV3, ResNet50, and DenseNet121.

3.5 Evaluation Metrics

To evaluate performance, we used metrics standard in medical image analysis, focusing on reducing false negatives critical for TB screening. Performance was assessed with Accuracy, Precision, Recall, Specificity, Negative Rate, and F1-score:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall \text{ (Sensitivity)} = \frac{TP}{TP + FN} \quad (3)$$

$$Specificity = \frac{TN}{TN + FP} \quad (4)$$

$$Negative \text{ Rate} = \frac{TN}{FN + TN} \quad (5)$$

$$F1\text{-Score} = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (6)$$

AUC-ROC was also used to evaluate discrimination ability. These metrics emphasize reducing false negatives, critical in TB screening.

4 Results and Discussion

4.1 Individual Model Performance

The performance of each individual CNN on the test set (at a threshold of 0.5) is detailed in Table 2. InceptionV3 showed the best standalone results, achieving 98% accuracy and a 97% F1-score, a success we link to its ability to extract features at multiple scales. DenseNet121 delivered well-rounded performance, with 92.3% accuracy and an 83% F1-score, thanks to its efficient reuse of features. ResNet50, however, demonstrated low sensitivity, with a recall of only 51%, leading to a low F1-score of 47% despite its high precision.

Table 2. Performance of CNNs and Ensemble.

Model	Acc (%)	Prec (%)	Rec (%)	F1 (%)	AUC
ResNet50	83.6	92.0	51.0	47.0	0.81
DenseNet121	92.3	94.0	78.0	83.0	0.92
InceptionV3	98.0	99.0	95.0	97.0	0.98
Ensemble	98.45	98.0	92.0	95.0	0.986

4.2 Ensemble Results and ROC Analysis

The ensemble improved recall and robustness compared to individual models. Figure 2 shows the confusion matrix with AUC=0.99.

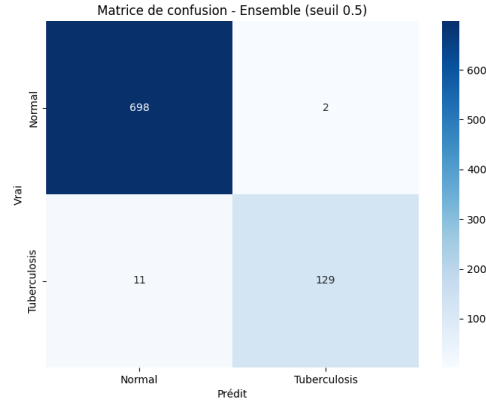


Fig. 2. Confusion matrix of ensemble model.

4.3 Threshold Analysis

An analysis of threshold sensitivity across 0.4–0.7 verified 0.5 as optimal for all models. InceptionV3 showed consistent performance across thresholds, while ResNet50’s recall was more sensitive to threshold changes. The ensemble displayed robustness to these variations, as shown in Figure 3.

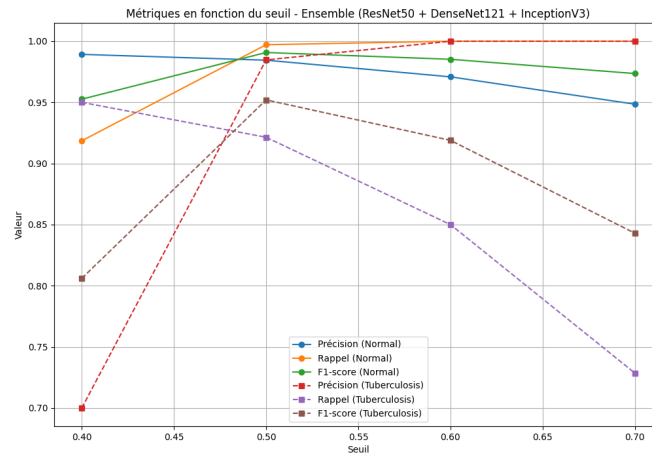


Fig. 3. Precision, Recall, and F1-score across thresholds.

5 Conclusion and Future Work

The TUBER-ENSEMBLE framework effectively shows that combining different CNN architectures is a powerful strategy for the automated detection of TB. With its soft voting mechanism, our ensemble reached an accuracy of 98.45%, surpassing both the individual models and other existing methods. The framework benefits from the distinct advantages of its components: the residual learning of ResNet50, the feature efficiency of DenseNet121, and the multi-scale analysis of InceptionV3. The main conclusions are that InceptionV3 delivers the best results for TB detection as a standalone model, that using ensemble methods markedly boosts robustness and the ability to generalize, and that the use of soft voting combined with threshold tuning improves the model's suitability for clinical use. The study's limitations include its validation on a single dataset and the computational demands that could hinder deployment on edge devices. Future research should focus on validating the model externally with data from diverse populations, developing lightweight model versions for use in low-resource settings, expanding the model to classify multiple diseases, and adding a lung segmentation step in preprocessing. This research lays the groundwork for AI-driven TB screening, a tool that could be particularly impactful for public health in high-prevalence areas with limited resources, where automated diagnostics can make a significant difference.

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